

Roll for the Galaxy: Mancala Dice Variant

This is the primary Roll prototype in this repository. It replaces the Dice Cup, roll, and assignment steps with a five-section mancala loop. The archived phase-battery rules remain in `docs/roll\_phase\_battery\_rules.md`.

Use the physical dice from the base box as mancala stones. Die faces do not matter. Die color is still identity for gained dice, lost dice, color bonuses, and end-game scoring.

Design Thesis

Dice stay in the mancala economy. They do not become workers, paid construction progress, or Goods.

The loop has five sections:

```
Explore -> Develop -> Settle -> Produce -> Ship -> Explore
```

Core economy:

- Sowing moves stones and selects a phase.
- Stones in a phase section set that phase's strength.
- Credits pay for Develop and Settle.
- Good markers track produced Goods on Worlds.
- Spent is off-board and holds overflow, lost dice, and newly gained dice that

cannot fit in their matching section.

This keeps the table state readable: the mancala board shows phase pressure, Credits show construction power, and Good markers show shipping inventory.

Components

Each player needs:

- one five-section mancala board;
- one off-board Spent area;

- Credits;
- VP chips;
- Good markers;
- their normal tableau, Development queue, and World queue.

Solo mode also needs:

- five Dummy Phase cards: Explore, Develop, Settle, Produce, Ship;
- one Dummy Goods marker;
- a shared VP pool.

Dice Colors

Blue	Explore identity
Brown	Develop identity
Red	Military / Settle identity
Green	Produce identity
Purple	Ship identity
White	Wild identity
Yellow	Alien identity

White and Yellow have no native entry section. When recovered from Spent, choose which section they enter.

Setup

For each player:

1. Place 2 Blue stones in Explore.
2. Place 2 Brown stones in Develop.
3. Place 2 Red stones in Settle.
4. Place 2 Green stones in Produce.
5. Place 2 Purple stones in Ship.
6. Place 2 White stones in Spent.
7. Start with 1 Credit.
8. Set up starting Worlds, starting Developments, and queued tiles normally for

this prototype.

Each section has a capacity of 6 stones. If no section can receive a stone during sowing, place that stone in Spent.

Round Structure

Each round:

1. Each player chooses one source: a board section, or one color group in Spent.
2. Each player sows those stones.
3. The final section sown selects that player's phase.
4. All selected phases occur in normal Roll phase order.
5. Your selected phase resolves at full section strength.
6. A phase selected only by another player gives you at most one matching-color assist.
7. Resolve Manage Empire.

There is no Citizenry and no dice assignment area.

Sowing

When sowing from a board section, pick up every stone in that section and sow clockwise into later sections, one stone per section.

When sowing from Spent, choose one color group and sow any number of those stones. Native colors enter their matching section:

```
Blue -> Explore
Brown -> Develop
Red -> Settle
Green -> Produce
Purple -> Ship
```

White and Yellow may enter any section.

Skip full sections. If every section is full, overflow stones go to Spent.

Match Bonuses

If the final stone lands in its matching section, mark one phase bonus for that section this round.

```
Blue in Explore      +1 Explore action
Brown in Develop     -1 Credit cost this Develop
Red in Settle        -1 Credit cost this Settle
Green in Produce     +1 Produce action
Purple in Ship       +1 VP on the first shipped Good this Ship
```

White and Yellow do not create match bonuses.

Phase Resolution

Stones stay on the mancala board when phases resolve.

Your own selected phase uses the full number of stones in that phase section.

Other players' selected phases are assists: you may resolve at most one action/discount, and only if you have the matching color in that section.

Blue in Explore	assists Explore
Brown in Develop	assists Develop
Red in Settle	assists non-Alien Settle
Yellow in Settle	assists Alien Settle
Green in Produce	assists Produce
Purple in Ship	assists Ship

White does not enable an assist by itself.

Explore

Explore actions equal stones in Explore plus the Explore match bonus if you selected Explore. An assisted Explore scouts one tile.

For each action:

1. Scout one new tile into the bottom of your shorter construction queue.
2. If the queues are tied, scout a Development.

Explore never gains Credits. It expands future build options without changing the current top queued Development or World.

Develop

Develop works on the top queued Development. If you selected Develop, use full section discounts. If Develop was selected only by another player, you may apply a one-Credit discount only if you have a Brown stone in Develop.

```
effective cost =
Development cost
- matching Brown stones in Develop
```

- White/Yellow stones in Develop
- Develop match bonus

Pay any number of Credits up to the remaining effective cost. Paid Credits stay on that queued Development as progress. If paid progress is still below the current effective cost, leave the Development queued and continue it in a later Develop phase.

Recalculate the section discount each time Develop occurs. Only paid Credits are permanent progress.

Settle

Settle works like Develop, but for the top queued World. If you selected Settle, use full section discounts. If Settle was selected only by another player, you may apply a one-Credit discount only if you have a Red stone in Settle.

Alien Worlds are the exception: you must have at least one Yellow stone in Settle to settle an Alien World. For an Alien World, only Yellow stones in Settle reduce the Credit cost; Red stones, White stones, and the Red Settle match bonus do not help. If Settle was selected only by another player, an Alien World assist requires Yellow in Settle and gives one Credit of discount.

```
effective cost =
World cost
- matching Red stones in Settle
- White/Yellow stones in Settle
- Settle match bonus
```

For an Alien World, use this instead:

```
effective cost =
Alien World cost
- Yellow stones in Settle
```

Pay any number of Credits up to the remaining effective cost. Paid Credits stay on that queued World as progress. If paid progress is still below the current effective cost, leave the World queued and continue it in a later Settle phase.

Recalculate the section discount each time Settle occurs. Only paid Credits are permanent progress.

Produce

Produce actions equal stones in Produce plus the Produce match bonus if you selected Produce. An assisted Produce places at most one Good and requires a Green stone in Produce.

Each action places one Good marker on an eligible empty production World in your tableau. The stone remains in Produce. The Good marker is not a die.

Ship

Ship actions equal stones in Ship if you selected Ship. An assisted Ship ships at most one Good and requires a Purple stone in Ship.

Each action ships one Good marker. If you have 2 or fewer Credits, trade the Good for Credits. Otherwise, consume it for VP chips. The Ship match bonus adds +1 VP to the first Good consumed during this Ship phase.

The shipped Good marker is removed. The Ship stone remains in Ship.

Manage Empire

During Manage Empire, spend 2 Credits to make one recovery sow from Spent. Repeat while you can pay and still have stones in Spent.

Credits do not become dice. Credits do not enter the mancala board. Credits pay construction costs and recover Spent stones.

Gaining And Losing Dice

When a tile grants a die:

1. Put it in its matching section if there is room.
2. Otherwise put it in Spent.

When a tile removes a die:

1. Remove from Spent first, avoiding colors the tile grants if possible.
2. If needed, remove from a board section.

Goods and construction queues are not dice.

Card Text Conversion

Use the following translation rules:

- Extra phase worker: add one temporary action or discount for that phase.
- Cost reduction: reduce the Credit cost of the matching Develop or Settle.
- Gain Credits: gain Credits normally, up to the Credit cap.
- Produce a Good: place a Good marker on an eligible World.
- Trade or Consume: use Ship actions and Good markers.
- Reassign a die: move an unused stone between board sections only if the text

clearly supports moving a die.

- Dice-count scoring: count physical dice in board sections and Spent only.

Ignore text that would place a die as lasting construction progress. This

variant tracks only Credit progress on queued Developments and Worlds.

Multiplayer Notes

All selected phases are shared, but not at full strength for everyone. If one

player selects Develop, every player may assist Develop, but only players with a

Brown stone in Develop get the one-discount assist. The selecting player resolves

Develop at full section strength.

The current prototype settings are:

```
Section capacity:      6
Starting stones:       2 per normal phase section
Starting white stones: 2 in Spent
Starting yellow stones: 0
Starting Credits:      1
Recovery sow cost:     2 Credits
VP pool:               8 per player
Tableau end trigger:   12 cards
```

Recent smoke simulation with four strategies averaged about 14-15 rounds with

matching-color assists.

Solo Mode

Solo mode uses the same player rules and adds a simple dummy phase deck.

Setup:

1. Build a five-card Dummy Phase deck: Explore, Develop, Settle, Produce, Ship.
2. Shuffle it.
3. Set the VP pool to 30.
4. Play up to 16 rounds.

Each round:

1. You sow and select one phase.
2. Draw 2 Dummy Phase cards.
3. Resolve your selected phase plus the dummy phases, in normal phase order; your phase is full strength and Dummy phases are matching-color assists.
4. Resolve the dummy effects.
5. Resolve Manage Empire.
6. If the Dummy deck cannot supply 2 cards, reshuffle all five Dummy Phase cards.

Dummy effects:

Explore	Dummy claims one Development and one World from the tile bag.
Develop	Dummy claims one Development.
Settle	Dummy claims one World.
Produce	Dummy Goods +1, max 4.
Ship	Drain 2 VP per Dummy Good, minimum 2, then Dummy Goods = 0.

Solo ends when any of these happen:

- you reach 12 tableau cards;
- the VP pool is empty;
- round 16 ends.

Solo Scenarios

Score scenarios:

Great	38+ VP
Triumphant	42+ VP
Epic	46+ VP

Named scenarios require at least 34 VP plus the listed mark:

Builder	7+ completed tiles
Developer	4+ Developments
Colonizer	6+ Worlds
Shipper	8+ VP chips
Expanded Workforce	17+ owned dice
Producer	4+ production Worlds
Diversified Economy	3+ distinct World colors
Phase Specialist	3+ color match bonuses
Credit Economy	18+ Credits spent
Logistics	2+ recovery sows
Phase Momentum	65+ phase actions

Owned dice means physical dice in your mancala sections plus Spent. Good markers and queued tiles do not count. Phase actions are actions produced by stones in resolved phase sections, including bonus actions from match bonuses.

Solo Campaigns

A campaign is four consecutive games. Each game must mark one campaign scenario not already marked. Win the campaign by marking all four scenarios.

Frontier Survey	Great, Colonizer, Diversified Economy, Phase Specialist
Core Worlds Renaissance	Triumphant, Developer, Builder, Credit Economy
Trade League	Triumphant, Producer, Shipper, Diversified Economy
Supply Lines	Great, Expanded Workforce, Logistics, Phase Momentum
Colonial Boom	Great, Colonizer, Producer, Logistics
Industrial Mobilization	Epic, Expanded Workforce, Credit Economy, Phase Momentum
Imperial Prestige	Epic, Builder, Shipper, Phase Specialist
Galactic Mastery	Epic, Credit Economy, Logistics, Phase Momentum

Campaign story hooks:

- Frontier Survey: chart a scattered sector and turn discoveries into a working

colony network.

- Core Worlds Renaissance: rebuild the home cluster through Developments,

infrastructure, and Credit discipline.

- Trade League: make production and shipping the center of galactic influence.
- Supply Lines: keep a wide empire supplied through workforce growth and

repeated recovery.

- Colonial Boom: expand fast, then make the new Worlds produce.
- Industrial Mobilization: push a large workforce and Credit economy into an

Epic-scale engine.

- Imperial Prestige: convert visible accomplishments into prestige before the

Dummy drains the sector.

- Galactic Mastery: a hard economy-control arc for strong solo play.

Current balanced-strategy tuning target:

Frontier Survey / Trade League / Colonial Boom: approachable campaign wins
Core Worlds / Supply Lines / Industrial Mobilization / Imperial Prestige / Mastery:
harder economy and prestige campaigns

Current Test Coverage

The automated tests cover:

- five-section setup and Spent;
- board sowing and Spent recovery;
- phase sharing;
- Develop/Settle Credit costs and section discounts;
- match bonuses;
- Produce Good markers without moving stones;
- Ship Good markers without moving stones;
- solo dummy phase selection;
- solo capacity counting and campaign definitions.